QIAGEN Genomic DNA Extraction Modified Protocol (plant and fungi as input)
Last updated: Aug 13, 2025

Prepare:

1. Prepare 2n mL **Buffer QF** in tubes and heat up to 50 ℃ in a water bath.
2. Put **glycogen** on ice .
3. Prepare at least half a liter of **liquid nitrogen**.
4. Prepare n mortars and n pestles.
5. Prepare a bucket of ice and put samples on ice.
6. Prepare DNA extraction Buffer*
7. Measure leaf tissue: 400 mg - 650 mg
8. Qiagen Genomic Tip (Mini kit)

Extraction Buffer

Reagent	Final Concentration	Volume	Stock Solution / Notes
EDTA (pH 8.0)	**20 mM**	**400 µL**	**0.5 M stock solution**
Tris (pH 8.0)	**10 mM**	**100 µL**	**1 M stock solution**
Triton X-100	**1% (v/v)**	**100 µL**	**-**
Guanidine HCl	**500 mM**	**625 µL**	**8 M stock solution**
NaCl	**200 mM**	**400 µL**	**5 M stock solution**
Viscozyme	**6 mg/mL**	**50 µL**	**1.2 g/mL, add just before use**
PVP	**2% (w/v)**	**0.2 g**	**Dissolve in 100 µL H_2O**
H_2O	**-**	**8325 µL**	**Adjust final volume to 2000 µL**

Total Volume	**-**	**10000 µL**	**-**

Homogenization (may not be necessary for fungal samples):

1. Clean a ladle with RNase and ethanol.
2. Pre-chill mortars and pestles with liquid nitrogen.
3. Homogenize the sample into a fine powder, but avoid over-grinding to prevent DNA degradation.
4. Transfer the powdered sample to a 15 mL tube and immediately add 10 mL of extraction buffer.

DNA Extraction:

- If skipping the homogenization step, directly add 10 mL of extraction buffer to the 15 mL tube containing the sample.
- Gently mix to suspend the sample in the buffer.

1. Incubate for 2 hrs at 37-40°C using a water bath. mixing occasionally by inversion.
2. Add RNase A to each tube to a final concentration of ~15-20 µg/mL, then incubate at 37-40°C for 20 minutes.
    - RNase A stock conc = 20 mg/mL, add 10 uL to each tube
3. Add Protease/proteinase K to each tube to a final concentration of ~0.8 mg/mL (recommended but ~0.4 mg/mL also works)
    - Incubate at 50°C for at least 2 hours, or overnight if needed, mixing occasionally by inversion.
    - Incubation time will vary depending on the sample.
    - Proteinase K stock conc. = 20 mg/uL add 100 uL
4. Centrifuge for 15 min at max speed using the large centrifuge (Eppendorf 5430).
5. filter the lysate through Miracloth into new 15 mL tubes. Use wide-bore pipe type tips to transfer lysate.
6. Use the cap of the 15 mL tube to secure the Miracloth. Centrifuge at low speed (1000-2500 rcf) to recover the liquid from the Miracloth. Remove the MiraCloth and then keep lysate on ice until usage.
7. Now you have to prime the filter from the Qiagen Blood & Cell Culture DNA Mini Kit (25). Get n x 15mL tubes and put filters using adapters onto the tubes.
8. Add 1mL Buffer QBT to each filter, and wait till empty by gravity flow.
9. Using wide-bore tips, transfer 3–5 mL of lysate (containing DNA) to the filter. Adjust the volume based on your sample to avoid clogging. Allow the lysate to pass through by gravity. Once filtration is complete, discard the flow-through

10. Wash the filters using 1mL of **Buffer QC** four times. Allow the buffer to pass through by gravity. Once filtration is complete, discard the flow-through
11. Dry the tip of the column using Kimwipes to remove any residual liquid.
12. Transfer the filters to new 15mL tubes.
13. Add 1 mL of 50 °C Buffer QF to each filter twice, allowing it to pass through by gravity each time. Do not discard the lysate, as it contains the DNA
14. Transfer the lysate into two 2mL **DNA LoBind tubes for each extraction using** wide-pore tips.
15. In each tube, add 0.7 mL **isopropanol** and 1 µL **glycogen**.
16. Precipitate the DNA by centrifuging at 8,000xg for 15 min at 4℃. <u>Make sure the opening of the centrifuge tubes are pointing towards the center.</u>
    - Prepare 2n mL fresh 70% **ethanol** and put on ice.
17. Mark the position of the DNA pellet on the side of the tube with a sharpie, then carefully remove the supernatant. The pellet can be difficult to see—holding the tube up to the light may help. If no pellet is visible, proceed to step 22; otherwise, continue to step 23
18. (Optional) Mix gently and incubate the samples at -20 ℃ for 1hr
    - Sample can also be left overnight
    - This only has to be done if DNA is not precipitating
19. Wipe out the mark on the tube.
20. Add 1mL of cold 70% **ethanol** and mix gently.
21. Precipitate the DNA by centrifuging at 8,000xg for 10 min at 4℃. <u>Make sure the opening of the centrifuge tubes are pointing towards the center.</u>
22. Mark the DNA pellet on the tubes and remove the supernatant carefully.
23. Air-dry till all ethanol evaporates.
24. Add 25µL **EB Buffer** per tube (50 µL per sample) and incubate overnight at 4℃ to allow DNA to resuspend in solution. Alternatively you can also incubate at 50℃ for 1 hr at 300 rpm, but some DNA will be lost. <u>Never freeze HMW DNA.</u>

<u>NOTE: Some EB buffers contain EDTA, which can interfere with downstream applications. Ideally, use EB buffer consisting of 10 mM Tris, pH 8.5, or opt for molecular-grade water instead.</u>

Use this table to record the number of operations that have been done during the extraction:

Steps \ Samples				
11. Buffer QC wash (4)				
14. Buffer QF elute (2)				